

Smart QR Code Attendance System & Security Architecture

Comprehensive Technical Specification, Database Schema, Workflow & Anti-Proxy Security Manual

Version: 2.4.0 (Enterprise)	Environment: Python Flask / MySQL	Generated: 25 September, 2026	Classification: Production Ready
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Executive Overview: CampusSync integrates a state-of-the-art dual-mode attendance tracking system (Manual Register + Live Dynamic QR Attendance). The QR engine is protected by a multi-layered security barrier comprising **GPS Geofencing (500m campus boundary)**, **Hardware Device Fingerprinting (1 Phone = 1 Student)**, **Single-Scan Replay Prevention**, and **Real-time Faculty Conflict Resolution**. This document details every schema change, technology library, operational step, and threat vector countermeasure.

1. System Architecture & Dual-Mode Operational Workflow

CampusSync eliminates conventional classroom roll-call bottlenecks while preserving faculty autonomy. When taking attendance for any lecture session, the faculty is presented with two explicit, mutually exclusive options:

- **Option 1: Manual Attendance Register:** The traditional student roster with present/absent radio toggles, search filters, and 'Mark All Present' convenience switches.
- **Option 2: Live QR Attendance:** A high-speed, projector-ready dynamic QR projection system enabling 60+ students to mark verified attendance concurrently within seconds.

Intelligent UI State Locking: Once the professor selects either option, the choice selection card is automatically removed from the viewport to eliminate UI clutter, locking the interface into that workflow.

End-to-End Operational Lifecycle:

Phase	Technical Operation & Logic
Step 1: Controlled Initialization (Generate Button)	Faculty selects lecture parameters and clicks 'Generate Attendance QR (10 Mins)' when ready to display on the projector. The backend generates a cryptographically random session token (e.g. CS-BCA101-A-8F2B1A) and sets is_qr_active = 1.
Step 2: Projection & Broadcast	A high-contrast 280x280 (or full-screen 360x360 projector mode) QR code is rendered containing the encrypted session token, campus geolocation anchor, and timestamp.
Step 3: Student Scanning	Enrolled students open their Student Portal (/student/scan). The portal verifies in real-time whether a lecture is active for their specific Semester & Division. If yes, the camera viewfinder activates.
Step 4: Multi-Tier Verification	When scanned, the student device submits GPS coordinates, hardware fingerprint hash, and token. The backend executes a 5-tier validation pipeline within 25 milliseconds.
Step 5: Real-Time Roster Sync	Upon verification, the student is marked 'Present'. The faculty dashboard auto-syncs via background polling (3-second cadence), incrementing attendance metrics, updating progress bars, and logging audit metadata.
Step 6: Session Termination	Upon lecture completion, faculty clicks 'Final Submit'. The backend immediately flips is_qr_active = 0, invalidating all outstanding tokens and preventing delayed or post-class proxy attempts.

2. Database Schema & Architecture (MySQL Database: campussync)

To support high-accuracy geofencing, hardware device binding, and anti-proxy conflict auditing, the MySQL relational database schema was enhanced with dedicated columns and a new security table. All tables use foreign-key integrity, indexed lookups, and transaction atomicity.

A. Table: college_settings (Campus Geofence Configuration)

Column Name	Data Type	Default / Value	Description & Security Purpose
campus_latitude	DECIMAL(10,8)	24.15953750	Exact GPS latitude anchor of college main building.
campus_longitude	DECIMAL(11,8)	72.40295313	Exact GPS longitude anchor of college main building.
campus_radius_meters	INT	500	Maximum allowable geofence radius perimeter in meters.
campus_plus_code	VARCHAR(100)	5C53+R58 Palanpur	Google Maps Open Location Code (OLC) reference.

B. Table: students (Hardware Device Fingerprint Binding)

Column Name	Data Type	Constraint	Description & Security Purpose
device_fingerprint	VARCHAR(128)	NULL / Indexed	Hardware-bound device hash. Enforces 1 Phone = 1 Student.
device_model	VARCHAR(100)	NULL	Operating system/device model (e.g., iPhone iOS, Samsung S23).
device_bound_at	DATETIME	NULL	Timestamp when student registered their personal mobile phone.
device_reset_allowed	TINYINT(1)	Default 0	Admin/Faculty permission flag allowing student to register a new phone.

C. Table: lecture_attendance_sessions (Dynamic QR Token Tracking)

Column Name	Data Type	Constraint	Description & Security Purpose
attendance_mode	VARCHAR(20)	Default 'Manual'	Mode used: 'Manual' or 'QR'.
qr_session_token	VARCHAR(64)	NULL / Unique	Cryptographically random token required for scan acceptance.
qr_session_expires_at	DATETIME	NULL	Token validity expiration boundary (Strict 10-Minute Window).
is_qr_active	TINYINT(1)	Default 0	Live toggle. 1 = Scanner enabled for class, 0 = Session closed.

D. Table: lecture_attendance_students (Scan Audit Trail)

Column Name	Data Type	Default	Description & Forensic Purpose
marked_method	VARCHAR(20)	'Manual'	Recorded as 'QR' or 'Manual' for transparency.
scanned_at	DATETIME	NULL	Exact millisecond timestamp when student camera captured QR.
device_fingerprint	VARCHAR(128)	NULL	Fingerprint of device used during this specific scan.
scan_latitude / longitude	DECIMAL(10,8)	NULL	GPS coordinates reported by student phone during submission.
distance_meters	FLOAT	NULL	Calculated distance from campus center (via Haversine formula).
is_verified	TINYINT(1)	0	1 = Verified by Geofence & Device Check; 0 = Pending/Unverified.

E. Table: attendance_security_alerts (Anti-Proxy Conflict Ledger)

Column Name	Data Type	Alert Triggers & Purpose
alert_type	VARCHAR(50)	OUT_OF_GEOFENCE, DUPLICATE_DEVICE, UNBOUND_DEVICE, TOKEN_MISMATCH.
session_id / student_id	INT / INT (FK)	Links alert to exact lecture session and offending student record.
conflicting_student_id	INT (FK)	Identifies original owner of device when proxy scan occurs.
faculty_action	VARCHAR(20)	'PENDING', 'APPROVED', 'REJECTED' — Faculty review disposition.
resolved_by_faculty_id	INT (FK)	Forensic trail of professor who approved or rejected the conflict.

3. Technology Stack & Software Libraries

CampusSync leverages industry-standard, lightweight, open-source libraries across the backend, frontend, and mobile-web scanner layers to maximize reliability, performance, and cross-device compatibility.

Library / Technology	Layer & Ecosystem	Exact Functional Role in System
Flask (Python 3.10+)	Backend Framework	RESTful API routing, Blueprint modularization, session management, and HTTP response handling.
Flask-SQLAlchemy & PyMySQL	Database ORM & Driver	Object-Relational Mapping, connection pooling, ACID transaction guarantees, and schema migrations.
Haversine Great-Circle Math	Backend Geofence Engine	Trigonometric spherical distance computation: calculates exact meter distance between mobile GPS & campus anchor.
Python secrets & hashlib	Security & Token Engine	Generates cryptographically unguessable hex salts (e.g. secrets.token_hex) and SHA-256 device hashes.
QRCode.js (Static 280x280)	Faculty Frontend	Client-side vector QR generation on HTML5 Canvas and SVG without external server round-trips.
HTML5-QRCode (v2.3.8)	Student Scanner Engine	Cross-platform mobile camera scanner; captures live video stream, decodes QR barcodes with 10 FPS rate.
W3C Geolocation API	Mobile Browser API	navigator.geolocation.getCurrentPosition with enableHighAccuracy:true to extract GPS coordinates.
HTML5 Web Storage (localStorage)	Device Identification	Persists client-side device signature hash across browser sessions to bind 1 Phone = 1 Student.
ReportLab 5.0.1	Document Engine	Generates pixel-perfect, secure audit PDFs with dynamic two-pass page numbering.

Haversine Spherical Distance Algorithm Implementation:

To verify whether a student is within the 500m campus boundary, the backend calculates the spherical great-circle distance:

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$$\Delta\phi = \text{radians}(\text{lat2} - \text{lat1}), \Delta\lambda = \text{radians}(\text{lon2} - \text{lon1})$$

$$a = \sin^2(\Delta\phi/2) + \cos(\phi1) \times \cos(\phi2) \times \sin^2(\Delta\lambda/2)$$

$$c = 2 \times \text{atan2}(\sqrt{a}, \sqrt{1 - a})$$

$$\text{Distance (m)} = \text{Earth\_Radius (6,371,000m)} \times c$$

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4. Security Questions, Threat Scenarios & Enterprise Solutions

A standard QR code scanner is vulnerable to basic exploitation: students photographing the projector screen, sending photos over WhatsApp to friends sleeping at home, or scanning on behalf of 5 classmates using a single phone. CampusSync solves every known vulnerability using our **Zero-Trust Anti-Proxy Architecture**:

Threat Vector & Vulnerability	Attack Scenario	CampusSync Solution & Enforcement
1. Remote Scanning / Proxy from Home Risk: Critical	A student in class takes a photo of the QR code and shares it on WhatsApp / Telegram with absent students at home.	GPS Geofence Barrier: Student GPS coordinates are extracted and validated on the backend. Any scan outside the 500m radius perimeter of campus anchor (24.15953750, 72.40295313) is instantly rejected with an OUT_OF_GEOFENCE security alert.
2. Multiple Scans from One Phone Risk: Critical	One student collects login credentials of 4 friends and logs in sequentially on their phone to scan the QR 5 times.	1-Device-1-Student Binding + Duplicate Device Guard: The first scan binds the hardware signature to the student. If the same device signature is detected for a second student in the same lecture, the system blocks the scan, flags a DUPLICATE_DEVICE alert, and triggers an immediate warning on the faculty screen.
3. Replay / Delayed Scan Attacks Risk: Medium	A student captures a QR code from a morning lecture and attempts to rescan it in the afternoon or evening.	Session & Token Expiry: When faculty closes attendance or generates a new QR, the backend sets is_qr_active = 0. Any scan submitted against an inactive session or mismatched salt is immediately rejected as expired.
4. GPS Spoofing / Mock Location Apps Risk: High	A tech-savvy student uses a fake GPS mocking app on Android to fake their location within the 500m college perimeter.	Multi-Factor Correlation: Spoofing GPS alone fails because the student still has to defeat Device Binding. Furthermore, faculty maintains an active live roster where real-time scans appear with student names and roll numbers. Suspicious anomalies can be marked Absent or audited with 1-click.
5. Lost / Changed Mobile Phone Risk: Operational	A student buys a new phone or broke their previous phone and cannot scan due to device mismatch.	Faculty / Admin Device Reset: College Admins or Faculty can toggle device_reset_allowed = 1 in the Student Master profile. On the student's next scan, their new phone is securely re-bound without technical support tickets.
6. Token Brute-Forcing Risk: Negligible	An outsider attempts to guess or automated-fuzz lecture tokens via API scripts.	Cryptographic Randomness: Tokens include high-entropy hexadecimal salts (16.7 million variations per session). Combined with active session filtering, brute-forcing within the 10-minute strict attendance window is mathematically infeasible.

5. Key Operational Questions Answered for Faculty & Management

Q1: What happens if a student does not own a smartphone or their battery dies?

CampusSync maintains 100% hybrid flexibility. Faculty can either click 'Mark' next to that student's name on the Live QR Roster, or simply switch to the Manual Register tab to check their radio button. Both modes feed into the same database tables seamlessly.

Q2: Why is the campus radius set to 500 meters?

Consumer smartphone GPS chips have an outdoor horizontal accuracy variance between 5 to 25 meters, which can degrade inside concrete multi-story academic buildings. A 500m radius covers the entire college perimeter (classrooms, labs, library, canteen) while strictly blocking proxy attempts from city residences, buses, or hostels located kilometers away.

Q3: Can faculty override a security alert if a student legitimately shared a device?

Yes. If a student's phone battery dies and they borrow a classmate's phone in front of the professor, an anti-proxy alert appears on the faculty screen. The professor can review the student's name and click 'Approve Exception', which records attendance with an official audit log.

Q4: Does the QR code refresh automatically during the lecture?

Faculty has a dedicated 'Regenerate QR' button. At any point during a 60-minute lecture, clicking Regenerate instantly issues a new token and updates the projector canvas, preventing any circulated photos from remaining valid.

Q5: Where is the attendance data permanently stored?

Every scan directly writes to `lecture_attendance_students` with `status='Present'` and `marked_method='QR'`. Simultaneously, the system automatically recalculates cumulative subject percentage statistics in `attendance_records`, which updates Student Dashboard meters, HNGU 75% eligibility warnings, and Dean/HOD reports in real time.

Developed & Validated By: CampusSync Engineering Team

Target Institution: CampusSync College (Palanpur)

Status: Production Deployed & Verified